# 17.1 - Approximately Normal Data Sets

Problem Solving, Math 1000

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## 1 Question

Often, in real-world observation, we might tend to see a pattern emerge relating the frequency of random events with the probabilities of each possible event. One of the most important patterns which might emerge in a bar graph of frequencies is called a bell curve, which looks like exactly what you might expect. If the bar graph is "perfectly" bell-shaped, we call this a normal distribution. If it is only roughly bell-shaped, we call this an approximately normal distribution. The emergence of this trend is all well and good, but the question remains: What can this tell us? How do normal distributions behave? Can we glean any useful information from this phenomena?

## 2 Example

The following depicts the frequency distribution of heads in 100 coin tosses and 10,000 repetitions of 100 coin tosses respectively. Notice that the data approaches a more perfect bell shape as the number of trials increases.

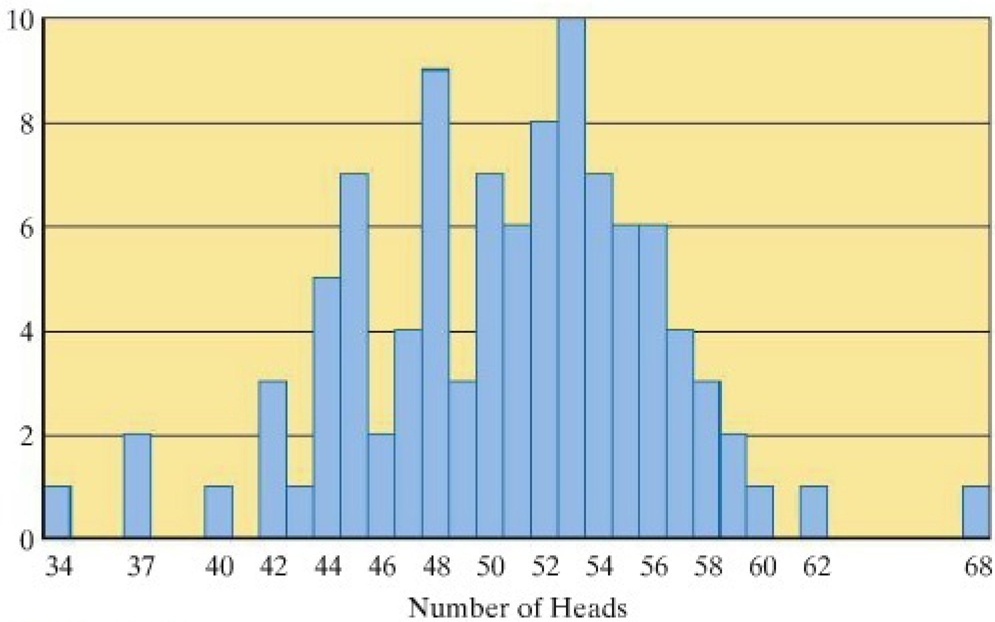


**FIGURE 17-4** Number of heads in 100 tosses (100 repetitions). *(Source:* Kerrich, J., *An Experimental Introduction to the Theory of Probability*, 1946.)

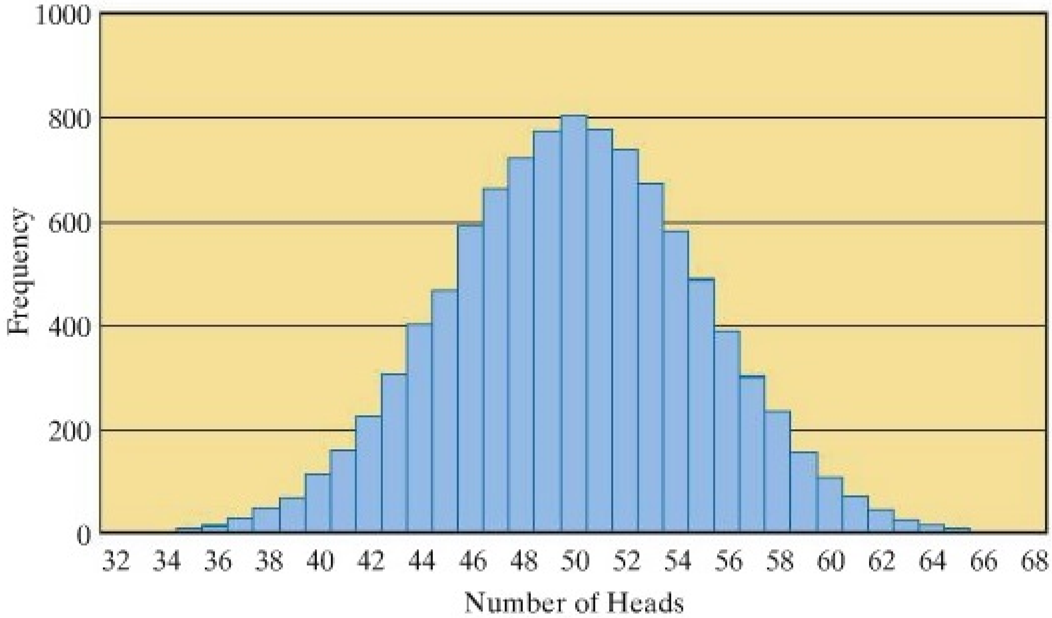


**FIGURE 17-5** Number of heads in 100 tosses (10,000 repetitions).
(*Source:* Computer simulation.)